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Wind load Analysis For Rolling Shutter

as per Indian Standards IS 875 Part III 1987

Project:	GODREJ & BOYCE MFG.CO. LTD
Client:	RBCC
Date:	13/07/2017
Height:	6000 mm
Width:	8000 mm
Slat Size:	75 mm
Slat Width:	15 mm

Wind Load Analysis as per IS

Design Wind Speed Vz= Vb * k1 * k2 * k3

where Vb = Basic Wind Speed in m / s. = 30 m/s

k1 = Risk Coefficient (clause 5.) = 1

k2 = THS Factor (clause 5.3 = 1

k3 = Topography Factor (clause 5.) = 1

Vz = Design Wind Speed = 30 m/s

Terrain Category: 2

(Open terrain with well-scattered obstructions having heights between 1.5m to 10m)

Class of Building: A(Structures having max. dimensions is less then 20m)

k1, k2 & k3 constants are derived from IS-875, based on location height of structure & geographical factor. These factors are required for tall structures, Where we need to consider varying pressure condition.

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Wind Load Analysis as per IS 875 - III

Design wind pressure = $0.6 \times V_z^2$

Where,

Pz = Design wind pressure

Vz = Design wind velocity

Wind Pressure, Pz = 0.6 x (30)2

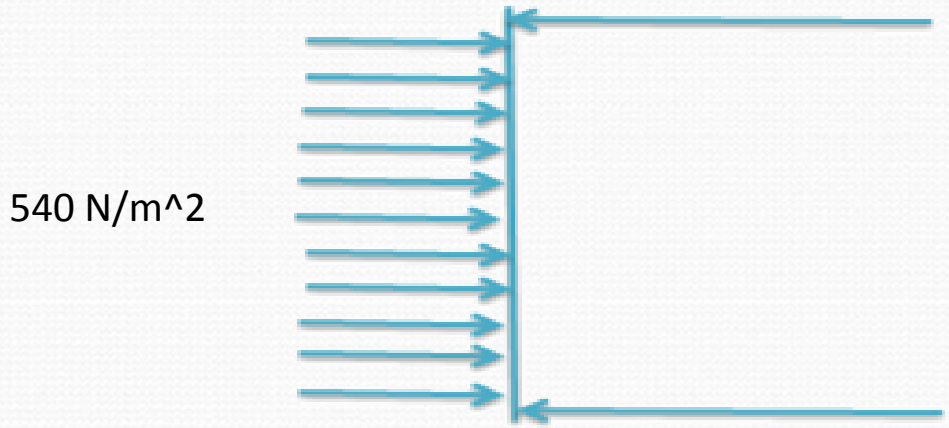
Wind Pressure, Pz = 540 N/m^2

Pz = 0.000540 N/mm^2

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Wind Load Analysis as per IS 875 - III

Assumed - Simply supported beam with uniform pressure.



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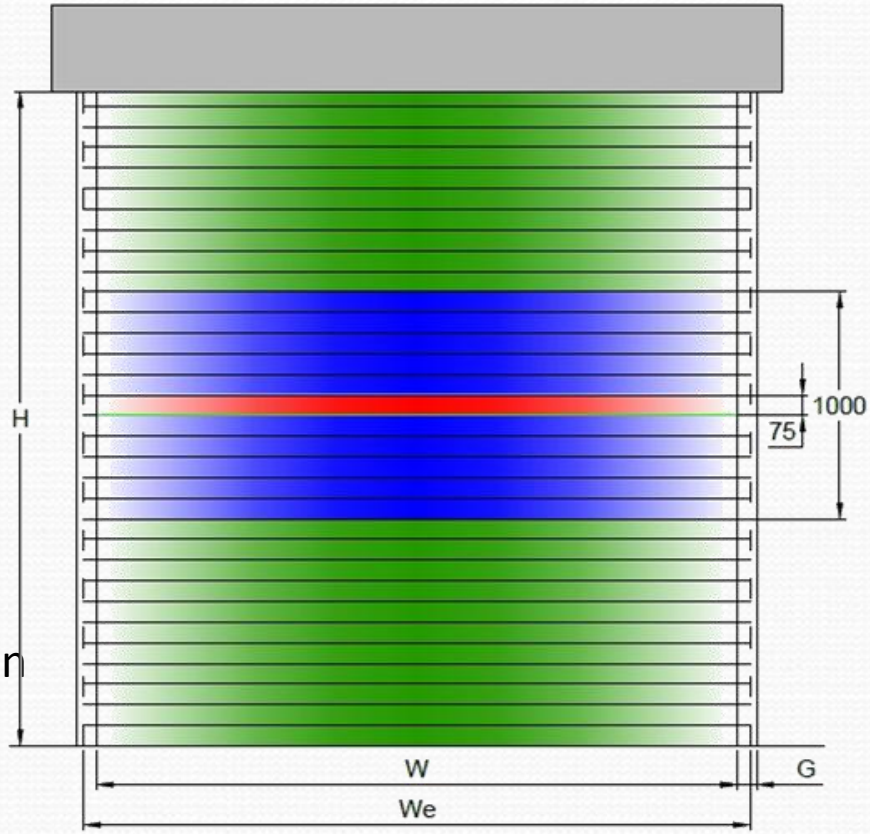
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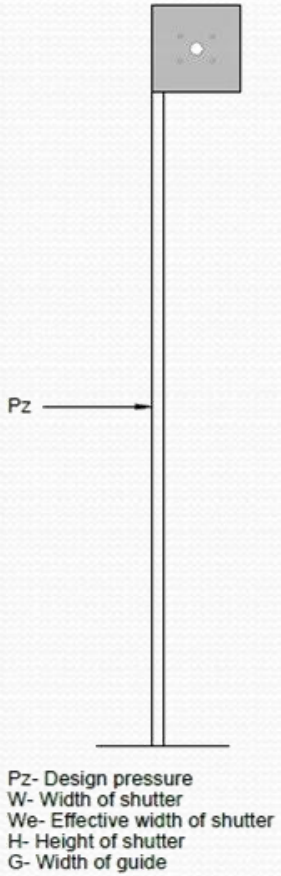
A)Bending Moment

B)Deflection

C)Maximum Stress

D)Side guides for curtain





Pz- Design pressure
W- Width of shutter
We- Effective width of shutter
H- Height of shutter
G- Width of guide

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A)Bending Moment

Door is simply supported on two opposite sides

Max B.M. $M = \alpha \times W \times L^2/8$

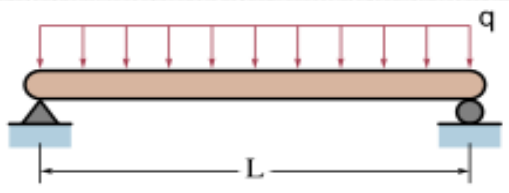
Where, W= UDL load

α= Bending moment coefficient as per IS 456:2000-Table26

B)Deflection

Max. deflection will be at the centre of slat

D max = (5x M x L^2) / (48 x E x I)



Where ,

M = Bending Moment of Slat

E = Modules of Elasticity for CRS = 210GPa (KN/mm2)

I = Moment of Inertia of slat (From CAD Data)

L = Clear width of Shutter

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C)Max Stress

Max. Stress= M x $\bar{y}$/ I

Where,

M=Bending moment of slat, $\bar{y}$= dist of side on y-y from CG , I= moment of inertia ,

Permissible Stress =248/FOS

=248/1.5

= 165 N/sq.mm

D)Side Guides for Curtain panel

Side Guides required = w/60

Where,

w= width of shutter

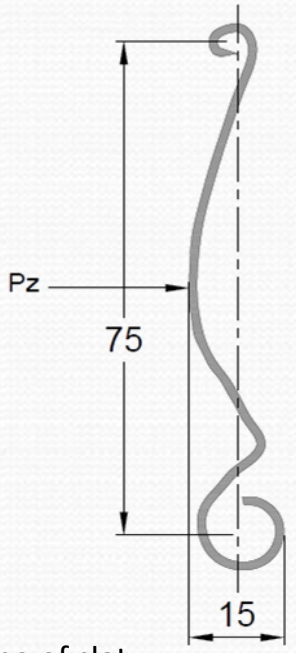
Design calculations for shutter of
Size: 8000mm(w) x 6000mm(H)

1. Wind Analysis for rolling shutter of size 8000mm(w) x 6000mm(h)

Design Wind Speed Vz= Vb * k1 * k2 * k3

where,

Vb = Basic Wind Speed in m / s.	=	30
k1 = Risk Coefficient (clause 5.4.1)	=	1
k2 = THS Factor (clause 5.3.2)	=	1
k3 = Topography Factor (clause 5.3.3)	=	1
Vz = Design wind Speed	=	30 m/s



Design wind pressure = 0.6x 30x30= 540 N/m2

UDL acting on Slat, W = Pz x A/L Where, A= Area of slat

= 0.00054 x (8000 x 75) / 8000

W = 0.0405 N/mm

Bending Moment $M = \alpha.(W \times L^2)/8$

= 0.079 x 0.0405 x 8000 x 8000 / 8

= 25596.00 N-mm

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Ø Assuming thickness of slat 1.2 mm & minimum height as 75mm & depth as 15mm.

(From CAD DATA) I= 1564 mm^4

E= modulus of elasticity for CRS= 210 GPA

= 210000 N/mm^2

Max. deflection = 5 x 25596 x 8000 x 8000 / (48 x 210000 x 1564)

= 519.55 mm

Permissible deflection = 8000/150

As per IS:800 = 53.333 mm

Max. stress = 25596x15/(2x1564)

= 122.74 N/mm^2

Permissible stress = (248 N/mm2) /1.5

= 165 N/mm^2

Guide side required = 8000/60 = 133.33

Hence we are using 150mm guide.

Note - As max. stress is less than permissible stress, hence no permanent deformation of slat will occur.

Thank You

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